

Giulio De Pasquale, PhD

SECURITY RESEARCHER

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Experience



Firmware Security Researcher

ROKU

Cambridge, United Kingdom

June 2025 - Present

Conduct comprehensive **firmware analysis** across Roku's entire device ecosystem, performing **static and dynamic analysis** on firmware and **bootloader security** spanning multiple architectures at enterprise scale.

Lead research initiatives focused on advancing **firmware security methodologies**, creating scalable analysis tools and developing new detection techniques for **embedded system vulnerabilities**.

🌐 <https://roku.com>



Security Researcher

APPLE

London, United Kingdom

October 2023 - June 2025

Worked on Dynamic Application Security Testing (**DAST**) infrastructure using **Golang**, implementing comprehensive vulnerability scanning capabilities across Apple's internal systems. Architected an extensible **plugin system** enabling security teams to develop custom detection modules and expand scanning functionality and led **CI/CD pipeline integration** and **DevOps practices**.

🌐 <https://apple.com>



Visiting PhD Student - SecLab

UNIVERSITY OF CALIFORNIA, SANTA BARBARA

Santa Barbara, California, USA

June 2022 - October 2023

Conducted research on applying **AI planning** in post-exploitation scenarios. This involved automatically deriving system facts and creating action chains for privilege escalation. This work was a component of the DARPA HARDEN program.

🌐 <https://seclab.cs.ucsb.edu> | 🧑🏫 Advised by Prof. Christopher Kruegel and Prof. Giovanni Vigna



Research Internship

APPLE

London, United Kingdom

May 2022 - June 2022

Internship in the Applied Research team part of Apple Information Security (AIS).

Developed an automated **firmware analysis** and discovered a zero-day vulnerability.

🌐 <https://apple.com> | 🧑🏫 Advised by Andre Protas



Visiting PhD Student

UNIVERSITY COLLEGE OF LONDON

London, United Kingdom

June 2021 - February 2024

Joined UCL's ISec research group to collaborate on ongoing research projects. Coordinated an ongoing series of fortnightly **hands-on seminars** on information security while focusing on solving CTF challenges.

🌐 <https://sec.cs.ucl.ac.uk>



Research Internship

TWEAG I/O

Remote

June 2021 - September 2021

Worked on improving the robustness of the type system of **Nickel**, a declarative configuration language. The project is written in **Rust**.

🌐 <https://github.com/tweag/nickel> | 🧑🏫 Advised by Dr. Yan Hamdaoui



Research Internship - NEXt Special Projects

MICROSOFT RESEARCH

Redmond, Washington, USA

June 2019 - September 2019

Researched **malware analysis** techniques to detect active threats in **ELF core** files as part of *Microsoft Project Freta*. The project has been developed in **Rust**.

🌐 <https://aka.ms/freta> | 🧑🏫 Advised by Mike Walker



Researched **deobfuscation** techniques for software protected by VM obfuscation and **fuzzing** methodologies targeting IoT devices.

 <https://seclab.cs.ucsb.edu> |  Advised by Prof. Christopher Kruegel and Prof. Giovanni Vigna

Publications

ChainReactor: Automated Privilege Escalation Chain Discovery via AI Planning

Philadelphia, Pennsylvania, USA

33RD USENIX SECURITY SYMPOSIUM (USENIX '24)

August 2024

ChainReactor is a research project that leverages **AI planning** to discover exploitation chains for **privilege escalation** on Unix systems. The project models the problem as a sequence of actions to achieve privilege escalation from initial access to a target system. **Awarded the Distinguished Artifact Award.**

• **Authors:**

Giulio De Pasquale, Ilya Grishchenko, Riccardo Iesari, Gabriel Pizarro, Lorenzo Cavallaro, Christopher Kruegel and Giovanni Vigna.

ROPfuscator: Robust Obfuscation with ROP

San Francisco, California, USA

17TH IEEE WORKSHOP ON OFFENSIVE TECHNOLOGIES (WOOT '23)

May 2023

Based on Return-Oriented Programming (ROP), ROPfuscator is a compiler-driven software **obfuscation pass**, developed to protect intellectual property in software from sophisticated man-at-the-end (MATE) attacks and reverse engineering attempts.

• **Authors:**

Giulio De Pasquale, Fukutomo Nakanishi, Daniele Ferla, Lorenzo Cavallaro.

DIANE: Identifying Fuzzing Triggers in Apps to Generate Under-constrained Inputs for IoT Devices

Online

42ND IEEE SYMPOSIUM ON SECURITY AND PRIVACY (S&P)

May 2021

DIANE is a tool that combines static and dynamic analysis to find **fuzzing triggers** and uses them to fuzz IoT devices automatically.

• **Authors:**

Nilo Redini, Andrea Continella, Dipanjan Das, **Giulio De Pasquale**, Noah Spahn, Aravind Machiry, Antonio Bianchi, Christopher Kruegel, Giovanni Vigna.

ShieldFS: A Self-healing, Ransomware-aware Filesystem

Los Angeles, California, USA

32ND ANNUAL COMPUTER SECURITY APPLICATIONS CONFERENCE (ACSAC) & BLACK HAT '17

December 2016

ShieldFS is an innovative solution to fight **ransomware** attacks. It automatically creates detection models that distinguish ransomware from benign processes at runtime on the base of the filesystem activity. ShieldFS adapts these models to the filesystem usage habits observed on the protected system.

• **Authors:**

Andrea Continella, Alessandro Guagnelli, Giovanni Zingaro, **Giulio De Pasquale**, Alessandro Barengi, Stefano Zanero, Federico Maggi.

Education



Ph.D Degree in Computer Security

London, United Kingdom

KING'S COLLEGE LONDON, DEPARTMENT OF INFORMATICS

November 2018 - February 2024

Specialized in **automated privilege escalation**, **program analysis**, **software obfuscation**, and **IoT firmware analysis research**.

Investigated methodologies to automate exploitation chain discovery and enhance resilience against reverse engineering and ransomware attacks.

 <https://s2lab.kcl.ac.uk> |  Advised by Prof. Lorenzo Cavallaro



B.Sc Degree in Engineering of Computing Systems

Milan, Italy

POLITECNICO DI MILANO

September 2011 - September 2017

Degree that covers all the fundamental topics in the IT area, such as Algorithms, Operating Systems, Databases, and Computer Architectures, combined with major Engineering subjects including Physics, Math, and Algebra.

Technical skills

Programming Python, Rust, Golang, Nix, C, Souffle, PDDL, TypeScript, Bash, C++, Java
Interests Reverse engineering, program analysis, system administration, declarative, low-level and systems' programming

Personal projects

Rustico

Rustico is a **Rust** cryptocurrency trading bot. It features a frontend developed with **Svelte and TailwindCSS**, and it is extendable with user-programmed trading strategies.

Pasticciotto

<https://github.com/peperunas/pasticciotto>

Pasticciotto is a polymorphic **Virtual Machine** that can be embedded in an application to protect proprietary code. It was developed for the **PoliCTF '17** as a reverse engineering challenge. The project is open source, and it is written in **C++**.

Injectopi

<https://github.com/peperunas/injectopi>

Injectopi is a set of tutorials that illustrates multiple **code injection** techniques in the Microsoft Windows environment. The project is open source, and it is written in **C**.

AESTracer

AESTracer is a **Microsoft Windows driver** which actively scans running processes for valid AES keyschedules. It has been included in a research project, ShieldFS, which was later published at ACSAC '16. The project was written in **C**.

Conferences contributions

Program Committee USENIX 2019, WOOT 2023, ESORICS 2023

Artifact Evaluation Committee USENIX 2021/2022/2023/2024, NDSS 2023, WOOT 2023

CTF Teams



mHACKeroni

ITALIAN CAPTURE THE FLAG (CTF) TEAM

Italy

May 2019 - PRESENT

mHACKeroni is composed by five CTF Italian teams. We qualified and placed 5th to **DEFCON 2019 Finals**.

- Involved in the organization of the **mHACKeCTF '20** competition.

<https://mhackeroni.it>



Phish 'n' Chips

KING'S COLLEGE LONDON CAPTURE THE FLAG (CTF) TEAM

London, United Kingdom

November 2018 - PRESENT

- Founder of the team.

<https://phishnchips.co.uk>



Shellphish

UCSB'S CAPTURE THE FLAG (CTF) TEAM

Santa Barbara, California, USA

January 2018 - PRESENT

- Involved in the organization of the **UCSB's iCTF 2018** competition.

<https://ictf.cs.ucsb.edu/>



Tower of Hanoi

POLITECNICO DI MILANO'S CAPTURE THE FLAG (CTF) TEAM

Milan, Italy

September 2014 - PRESENT

- Involved in the organization of the **PoliCTF 2015** competition held during DIMVA 2015 in Milan.

<https://polictf.it>